

Proposed List of Projects for Student Registration

1. Corporate Network Infrastructure Design

Business Use Case:

A growing company needs a **scalable network infrastructure** to support its operations and future expansion.

Guided Steps:

1. Design a **network topology** in Cisco Packet Tracer with routers, switches, and end devices.
2. Configure **subnetting** for different departments (HR, IT, Finance, etc.).
3. Set up **static and dynamic routing** for inter-department communication.
4. Implement **Quality of Service (QoS)** for VoIP and video conferencing.
5. Test network connectivity and performance.

Business Plan:

- **Service:** Provide **customized network design services** for businesses.
- **Revenue Model:** Charge for **network planning, setup, and performance optimization**.
- **Target Market:** Growing companies **expanding IT infrastructure**.
- **Growth Strategy:** Offer **network management as a service (NaaS)**. Tools

Used:

Cisco Packet Tracer

2. VoIP Implementation for Business Communication

Business Use Case:

A business wants to replace **traditional phone lines** with a **cost-effective VoIP system** for internal and external communication.

Guided Steps:

1. Set up a **VoIP network** using Packet Tracer.
2. Configure IP phones and a **call manager**.
3. Set up **SIP trunking** for external calls.

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4. Optimize **bandwidth allocation** for VoIP traffic.
5. Test call quality using simulated business scenarios.

Business Plan:

- **Service:** Offer **VoIP setup and optimization** for businesses.
- **Revenue Model:** Charge for **installation, configuration, and VoIP troubleshooting**.
- **Target Market:** Call centers, remote work companies, and **customer support teams**.
- **Growth Strategy:** Provide **hosted VoIP solutions** with **maintenance plans**. **Tools**

Used:

Cisco Packet Tracer, Asterisk/FreePBX

3. Office Network Setup s Performance Testing

Business Use Case:

A business moves to a new office and needs a **high-performance, reliable network**.

Guided Steps:

1. Design a network with **wired and wireless connections**.
2. Assign **IP addresses** using DHCP.
3. Configure **trunking and VLANs** for department segmentation.
4. Use **Wireshark** to analyze network performance.
5. Provide recommendations for **improving speed and reliability**.

Business Plan:

- **Service:** Offer **business network installation s optimization**.
- **Revenue Model:** Charge for **network deployment and performance audits**.
- **Target Market:** Small and medium-sized businesses (SMBs).
- **Growth Strategy:** Expand services to **managed network solutions**. **Tools**

Used:

Cisco Packet Tracer, Wireshark

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4. Enterprise Network Traffic Analysis

Business Use Case:

A company wants to **understand and optimize** its network traffic to improve business operations.

Guided Steps:

1. Capture live traffic using **Wireshark**.
2. Analyze bandwidth usage by departments.
3. Identify **bottlenecks** and **high-traffic applications**.
4. Optimize network paths using **load balancing**.
5. Create a report with **improvement recommendations**.

Business Plan:

- **Service:** Offer **network traffic analysis and optimization**.
- **Revenue Model:** Charge businesses for **traffic monitoring s performance reports**.
- **Target Market:** Enterprises with **high data traffic** (e.g., media companies, cloud services).
- **Growth Strategy:** Develop an **automated network monitoring dashboard**. Tools

Used:

Wireshark, NetFlow Analyzer

5. DNS and DHCP Implementation for Business Networks

Business Use Case:

A company needs **centralized management of IP addresses and domain names**. **Guided Steps:**

1. Set up a **DHCP server** to dynamically assign IP addresses.
2. Configure a **DNS server** to resolve business domain names.
3. Test **network-wide connectivity** using nslookup and ping.
4. **Optimize lease time settings** for efficient IP allocation.
5. Provide documentation for **network administrators**.

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Business Plan:

- **Service:** Provide DNS/DHCP configuration and management.
- **Revenue Model:** Charge for server setup, monitoring, and maintenance.
- **Target Market:** Offices with growing network infrastructures.
- **Growth Strategy:** Offer cloud-based DNS and DHCP services.

Tools Used:

Cisco Packet Tracer, Windows/Linux DHCP C DNS servers

6. Multi-Branch Business Network Configuration

Business Use Case:

A company with multiple locations needs a centralized network for secure and efficient communication.

Guided Steps:

1. Design a **WAN topology** in Packet Tracer.
2. Configure **VPN connections** between branches.
3. Implement **dynamic routing protocols (OSPF, BGP)** for efficient data transfer.
4. Test **inter-branch connectivity** and failover scenarios.
5. Provide a **scalability plan** for future expansion.

Business Plan:

- **Service:** Offer multi-branch network integration.
- **Revenue Model:** Charge businesses for setup, maintenance, and network expansion.
- **Target Market:** Companies with regional or international offices.
- **Growth Strategy:** Provide SD-WAN as a service for scalability.

Tools Used:

Cisco Packet Tracer

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7. Web Hosting Business (Web Server Project)

Project Description:

Create a Python-based web server that hosts static HTML pages. The server will listen for HTTP requests and serve webpages accordingly.

Implementation Steps:

1. Set up a basic HTTP server using Python's socket module.
2. Configure the server to handle multiple page requests.
3. Implement error handling (404 Not Found, 500 Internal Server Error).
4. Test with a browser by serving an HTML page.

Business Model:

- Offer low-cost web hosting for small businesses that need simple websites.
- Charge a monthly fee for hosting and maintenance services.
- Upsell custom website development and security monitoring services.

Tools:

Python (socket) , Basic HTML files , Web browser for testing

8. Business Proxy Server (Secure Internet Access)

Project Description:

Develop a proxy server that acts as an intermediary between client devices and the internet, improving security and speed.

Implementation Steps:

1. Use Python (socket) to intercept and forward requests.
2. Implement caching for faster load times.
3. Add content filtering (e.g., block malicious websites).
4. Test by routing browser traffic through the proxy.

Business Model:

- Offer secure internet access services to schools and offices.
- Sell subscription-based access to a private proxy server.

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- Monetize content filtering for parental control services.

Tools:

Python (socket, Flask) , Web browsers, Wireshark (to analyze traffic)

9. Network Analysis Business (Packet Sniffer Project)

Project Description:

Build a packet sniffer that captures and analyzes network traffic, identifying potential security threats.

Implementation Steps:

1. Use Scapy to capture network packets.
2. Extract source/destination IPs, protocols, and payloads.
3. Implement filtering for specific traffic types (e.g., HTTP, FTP).
4. Generate a report summarizing captured traffic.

Business Model:

- Offer network monitoring services for small businesses.
- Sell traffic analysis reports to IT departments.
- Charge for monthly network audits and troubleshooting.

Tools:

Python (Scapy, socket), Wireshark for comparison , Report generation tools (Excel, Python Matplotlib)

10. FTP Hosting Business

Project Description:

Develop a simple FTP server where users can upload and download files remotely.

Implementation Steps:

1. Create an FTP server using Python's ftplib.
2. Implement file upload/download functionality.
3. Add user authentication for security.
4. Allow multiple concurrent connections.

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Business Model:

- Provide remote storage solutions for small businesses.
- Charge a monthly fee for secure file storage.
- Offer backup and recovery services as an upsell.

Tools:

Python (ftplib, socket) , File system storage, Wireshark for monitoring

11. Secure Business Chat System

Project Description:

Develop a TCP-based chat system where multiple clients can communicate in real time.

Implementation Steps:

1. Create a TCP server using Python's socket.
2. Allow multiple clients to connect.
3. Implement user authentication and message encryption.
4. Add a logging system to store chat history.

Business Model:

- Sell secure messaging solutions to small businesses.
- Charge subscription fees for hosted chat services.
- Upsell end-to-end encryption and secure file sharing.

Tools:

Python (socket, threading), SQLite (for chat history), OpenSSL (for encryption)

12. WiFi Network Audit Service

Project Description:

Develop a WiFi scanning tool to analyze nearby networks and detect potential security risks.

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Implementation Steps:

1. Use iwlist (Linux) or netsh (Windows) to list nearby WiFi networks.
2. Extract SSID, signal strength, security type, and frequency.
3. Display data in a structured table.

Business Model:

- Offer WiFi security assessments to businesses.
- Sell consulting services for optimizing network performance.
- Charge a fee for penetration testing and WiFi hardening services.

Tools:

Python (subprocess for system commands), Wireshark for deeper analysis , tabulate Python package for output formatting

13. Network Vulnerability Scanning Business (Port Scanner)

Project Description:

Develop a port scanner that identifies open ports on target devices.

Implementation Steps:

1. Use Python's socket to scan specified IP ranges.
2. Identify open and closed ports.
3. Implement stealth scanning techniques.
4. Generate a report with security recommendations.

Business Model:

- Offer network vulnerability scanning for companies.
- Sell penetration testing reports to IT departments.
- Provide consulting services for firewall and server security.

Tools:

Python (socket, argparse), Wireshark for verification, Report generation tools (Excel, JSON logs)

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14. Secure Email Services (SMTP Email Client)

Project Description:

Develop a simple SMTP email client that sends emails without using smtplib.

Implementation Steps:

1. Use Python's socket to establish an SMTP connection.
2. Implement SMTP commands manually (HELO, MAIL FROM, RCPT TO).
3. Send email messages over the network.
4. Ensure spam protection by testing with real email providers.

Business Model:

- Sell custom email hosting services to small businesses.
- Offer secure email encryption as a paid feature.
- Provide email marketing tools and automation services.

Tools:

Python (socket), Email servers (Gmail SMTP, Office365 SMTP), Encryption libraries (OpenSSL)